

Why CFR converges

The complete proof, built from scratch — no prior knowledge assumed

Third volume of the PokerTrainer tutorial series

PokerTrainer documentation

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Abstract

The first volume of this series (*Deep CFR, explained*) *used* the fact that CFR self-play converges to a Nash equilibrium. This volume *proves* it. The proof is genuinely beautiful and needs nothing beyond careful algebra: three independent ingredients — a $\sqrt{T}$ bound for regret matching, a six-line “folk theorem” linking regret to equilibria, and Zinkevich’s decomposition of global regret into per-infoset regrets — snap together into the final theorem. Every step is motivated before it is taken, every symbol is introduced before it is used, and a real numerical simulation shows the theorem doing its work.

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1 What we are going to prove, and the plan

Here is the claim, first in plain words, then precisely.

In plain words: two players repeatedly play a zero-sum game (such as heads-up poker) against each other. At every decision point, each player runs the same simple bookkeeping rule — regret matching. Nobody is trying to “converge” to anything; each player just keeps updating its regrets. And yet, the **average** of all the strategies each player has used becomes arbitrarily close to a Nash equilibrium — the unexploitable, “GTO” strategy — with an error that shrinks like $1/\sqrt{T}$ after T iterations.

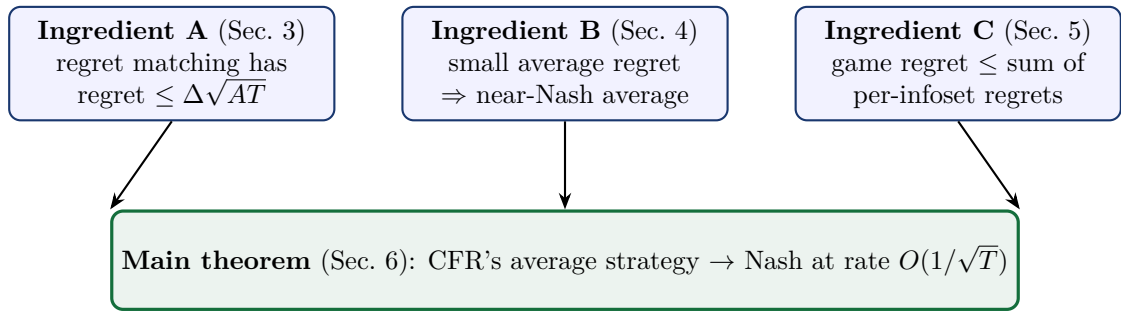
Main theorem (Zinkevich et al. 2007, informal form)

After T iterations of CFR self-play in a two-player zero-sum game with perfect recall, the pair of average strategies $(\bar{\sigma}_1, \bar{\sigma}_2)$ is an ε -Nash equilibrium with

$$\varepsilon \leq \frac{2 \Delta |\mathcal{I}| \sqrt{A}}{\sqrt{T}},$$

where Δ is the payoff range of the game, $|\mathcal{I}|$ the number of information sets of one player, and A the maximum number of actions at any of them. In particular $\varepsilon \rightarrow 0$: average CFR self-play converges to GTO.

The proof has three independent ingredients, each interesting on its own:



Notation we will use throughout (every symbol is re-introduced where it first matters):

symbol	meaning	first used
a, A	an action; the number of available actions	Sec. 2
t, T	current round; total number of rounds	Sec. 2
$u_t(a)$	payoff action a would have earned in round t	Sec. 2
σ_t	our mixed strategy (probabilities) in round t	Sec. 2
Δ	payoff range: $\max u - \min u$	Sec. 2
$r_t(a), R_T(a)$	instantaneous / cumulative regret of a	Sec. 2
x^+	positive part: $\max(x, 0)$	Sec. 3
$\bar{\sigma}$	time-averaged strategy	Sec. 4
$I, \mathcal{I}$	an information set; the set of all of them	Sec. 5
$v(I, a)$	counterfactual value of a at I	Sec. 5

2 The arena: one repeated decision

Forget game trees for now. The entire machinery is easiest to understand in the simplest possible setting, and ingredient A lives there entirely.

You face the *same* decision over and over: rounds $t = 1, 2, \dots, T$. In each round you must choose among A actions, and you are allowed to randomize: you pick a probability vector σ_t (your *mixed*

strategy) and play a draw from it. **After** the round, the environment reveals what every action would have paid this round: a vector u_t , with $u_t(a)$ the payoff of action a . Crucially, we make *no assumption* on how u_t is produced — it may depend on your past play, it may be chosen by an adversary who knows your algorithm. The only structure is a bounded range: all payoffs live in an interval of width Δ .

Your *expected* payoff in round t is $\langle \sigma_t, u_t \rangle = \sum_a \sigma_t(a) u_t(a)$.

Regret. The *instantaneous regret* of action a in round t is how much better a would have done than what you actually did:

$$r_t(a) = u_t(a) - \langle \sigma_t, u_t \rangle.$$

The *cumulative regret* after T rounds is $R_T(a) = \sum_{t=1}^T r_t(a)$, and your (*external*) regret is $R_T = \max_a R_T(a)$: how much you lost compared to the best *single fixed action* chosen with hindsight.

Note two easy facts we will use constantly. First, since both $u_t(a)$ and $\langle \sigma_t, u_t \rangle$ lie in the same interval of width Δ , every instantaneous regret is bounded: $|r_t(a)| \leq \Delta$. Second, the strategy-weighted average of the instantaneous regrets is *always zero*:

$$\sum_a \sigma_t(a) r_t(a) = \sum_a \sigma_t(a) u_t(a) - \langle \sigma_t, u_t \rangle \cdot \underbrace{\sum_a \sigma_t(a)}_{=1} = 0.$$

Regret is a *relative* quantity — you cannot regret everything at once.

Intuition

Why is “low regret against the best fixed action” the right goal, rather than “high payoff”? Because against an unknown, possibly adversarial environment, payoff itself is not controllable — the environment can always pay you nothing. What *is* controllable is the gap to the best hindsight action. And as Section 4 shows, in a two-player zero-sum game this modest goal is secretly equivalent to finding an equilibrium.

Regret matching (Hart & Mas-Colell 2000) is the bookkeeping rule from volume one: play each action proportionally to its positive cumulative regret, $\sigma_{t+1}(a) = R_t^+(a) / \sum_b R_t^+(b)$, and play anything (say uniform) if no regret is positive. Our question for the next section: *how big can R_T get under regret matching?*

3 Ingredient A: regret matching’s regret grows only like $\sqrt{T}$

Theorem A (Hart & Mas-Colell 2000)

Against *any* sequence of payoff vectors with range Δ , regret matching guarantees

$$R_T = \max_a R_T(a) \leq \Delta \sqrt{AT}.$$

Hence the *average* regret $R_T/T \leq \Delta \sqrt{A/T}$ vanishes as $T \rightarrow \infty$.

A bound that holds against any adversary, for such a naive rule, should feel surprising. The proof is three small steps. The object to watch is the **potential**

$$\Phi_T = \sum_a (R_T^+(a))^2,$$

the squared length of the *clipped* regret vector. We will show Φ can only grow by a bounded amount per round — which forces every individual regret to stay $O(\sqrt{T})$.

3.1 Step 1: the key cancellation

Lemma 1 (the regret-matching identity)

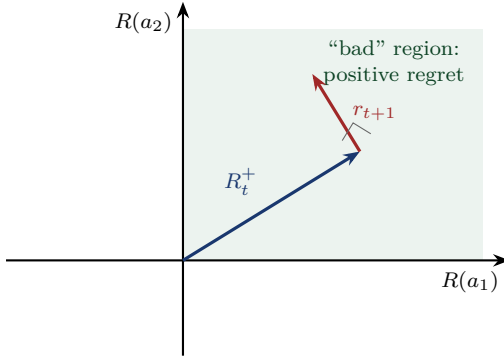
If σ_{t+1} is chosen by regret matching from R_t , then

$$\sum_a R_t^+(a) r_{t+1}(a) = 0.$$

Proof. If all regrets are ≤ 0 the left side is trivially zero (every $R_t^+(a) = 0$). Otherwise let $Z = \sum_b R_t^+(b) > 0$, so $\sigma_{t+1}(a) = R_t^+(a)/Z$. Then

$$\sum_a R_t^+(a) r_{t+1}(a) = \sum_a R_t^+(a) u_{t+1}(a) - \left(\sum_a R_t^+(a) \right) \langle \sigma_{t+1}, u_{t+1} \rangle.$$

The second term is $Z \cdot \sum_a \frac{R_t^+(a)}{Z} u_{t+1}(a) = \sum_a R_t^+(a) u_{t+1}(a)$ — identical to the first. They cancel. $\square$



Lemma 1, geometrically: the next regret increment is orthogonal to the current clipped regret vector. The update can move the vector sideways but never push it further out along its own direction — so its length grows like a random walk ($\sqrt{T}$), not like a straight march (T).

3.2 Step 2: clipping never hurts

Lemma 2 (clipping inequality)

For any real numbers x, y : $((x + y)^+)^2 \leq (x^+ + y)^2$.

Proof. If $x + y \leq 0$ the left side is 0 and the right side is a square, hence ≥ 0 . If $x + y > 0$, then since $x^+ \geq x$ we have $x^+ + y \geq x + y > 0$, and squaring preserves order on positives. $\square$

3.3 Step 3: the potential argument

Now bound the growth of Φ . For each action a , apply Lemma 2 with $x = R_t(a)$, $y = r_{t+1}(a)$, and sum over a :

$$\Phi_{t+1} = \sum_a ((R_t(a) + r_{t+1}(a))^+)^2 \leq \sum_a (R_t^+(a) + r_{t+1}(a))^2 = \Phi_t + 2 \underbrace{\sum_a R_t^+(a) r_{t+1}(a)}_{=0 \text{ (Lemma 1)}} + \sum_a r_{t+1}(a)^2.$$

The middle term — the only one that could grow linearly — is annihilated by Lemma 1. The last term is at most $A\Delta^2$ because each $|r_{t+1}(a)| \leq \Delta$. So

$$\Phi_{t+1} \leq \Phi_t + A\Delta^2 \quad \implies \quad \Phi_T \leq T A \Delta^2.$$

Finally, a single squared coordinate is at most the whole sum of squares:

$$R_T(a) \leq R_T^+(a) \leq \sqrt{\Phi_T} \leq \Delta \sqrt{AT} \quad \text{for every } a. \quad \blacksquare$$

Intuition

The whole theorem is the picture under Lemma 1. The clipped regret vector performs a dance in which every step is *perpendicular* to its current position (plus a bounded-length correction from clipping). Perpendicular steps of bounded size grow a vector’s *squared* length by a constant per step — the same reason a random walk wanders only $\sqrt{T}$ from home. “ $\sqrt{T}$ regret” is not a coincidence of regret matching; it is the signature of any algorithm that manages to keep its updates orthogonal to its accumulated mistakes.

3.4 Watching the theorem work

The table below is a *real* simulation (seed 7) of two regret matchers playing rock–paper–scissors against each other with sampled actions ($\Delta = 2$, $A = 3$, so Theorem A promises $R_T \leq 2\sqrt{3T}$):

T	$\max_a R_T(a)$	bound $2\sqrt{3T}$	R_T/T	avg. strategy (R, P, S)
10	3.3	11.0	0.327	(0.103, 0.730, 0.167)
100	9.9	34.6	0.099	(0.346, 0.399, 0.255)
1 000	28.3	109.5	0.028	(0.355, 0.316, 0.329)
10 000	92.0	346.4	0.009	(0.347, 0.330, 0.323)
100 000	351.8	1095.4	0.004	(0.336, 0.328, 0.336)

Three things to see. (1) Regret grows, but each $\times 10$ in T multiplies it by only $\approx 3.2 \approx \sqrt{10}$ — the square-root law, well inside the bound. (2) *Average* regret melts away. (3) The average strategy approaches $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, the Nash equilibrium of RPS — even though at $T=10$ the player was a wild Paper-spammer. That last observation is not explained by Theorem A at all. It is the subject of ingredient B.

4 Ingredient B: low regret $\Rightarrow$ near-equilibrium (the folk theorem)

Now let the “environment” be another player. Two players repeatedly play a **zero-sum** game: player 1 picks mixed strategy σ_1^t , player 2 picks σ_2^t , and player 1’s expected payoff is the bilinear $u(\sigma_1^t, \sigma_2^t)$, with player 2 receiving its negation. Each player measures its own regret exactly as in Section 2 (for player 1 in round t , the payoff vector is $u_t(a) = u(a, \sigma_2^t)$ — what each of my actions would have earned against what the opponent actually played).

ε -Nash equilibrium. A profile $(\bar{\sigma}_1, \bar{\sigma}_2)$ such that neither player can gain more than ε by deviating unilaterally: $u(\sigma'_1, \bar{\sigma}_2) \leq u(\bar{\sigma}_1, \bar{\sigma}_2) + \varepsilon$ for every σ'_1 , and symmetrically for player 2. The smallest such ε is the profile’s *exploitability*: how much a perfect counter-strategy could win above the equilibrium value.

Average strategy. $\bar{\sigma}_i = \frac{1}{T} \sum_{t=1}^T \sigma_i^t$ — the mixture that plays a uniformly random past round’s strategy. The single property we need is linearity of expected payoff in each argument:

$$\frac{1}{T} \sum_{t=1}^T u(\sigma'_1, \sigma_2^t) = u\left(\sigma'_1, \frac{1}{T} \sum_{t=1}^T \sigma_2^t\right) = u(\sigma'_1, \bar{\sigma}_2).$$

Playing against the average of my opponent’s past selves is the same as playing against each past self with probability $1/T$.

Theorem B (the folk theorem of online learning)

If after T rounds player 1's average regret is at most ε_1 and player 2's is at most ε_2 , then $(\bar{\sigma}_1, \bar{\sigma}_2)$ is an $(\varepsilon_1 + \varepsilon_2)$ -Nash equilibrium.

Proof. Let $\bar{v} = \frac{1}{T} \sum_t u(\sigma_1^t, \sigma_2^t)$ be the average realized payoff (to player 1). Player 1's regret guarantee says: for *every* alternative σ'_1 ,

$$\frac{1}{T} \sum_t u(\sigma'_1, \sigma_2^t) - \bar{v} \leq \varepsilon_1 \quad \xRightarrow{\text{linearity}} \quad u(\sigma'_1, \bar{\sigma}_2) \leq \bar{v} + \varepsilon_1. \quad (*)$$

Player 2's guarantee, written in player-1 payoffs (player 2 *earns* $-u$, so low regret for 2 means no σ'_2 pushes u much *below* $\bar{v}$):

$$u(\bar{\sigma}_1, \sigma'_2) \geq \bar{v} - \varepsilon_2 \quad \text{for every } \sigma'_2. \quad (**)$$

Now read the two lines together. Setting $\sigma'_2 = \bar{\sigma}_2$ in (**): $u(\bar{\sigma}_1, \bar{\sigma}_2) \geq \bar{v} - \varepsilon_2$. Hence for every deviation σ'_1 :

$$u(\sigma'_1, \bar{\sigma}_2) \stackrel{(*)}{\leq} \bar{v} + \varepsilon_1 \leq u(\bar{\sigma}_1, \bar{\sigma}_2) + \varepsilon_1 + \varepsilon_2.$$

Player 1 cannot gain more than $\varepsilon_1 + \varepsilon_2$ by deviating; the argument for player 2 is the mirror image. $\square$

Intuition

Six lines, and they explain the third column of our RPS table. Each player's regret guarantee constrains the *opponent's average*: “my regret is small” literally means “no fixed strategy beats the average of what my opponent has been doing by much”. Two such statements, one per player, pin the average profile to the equilibrium from both sides. Note also *what failed*: nothing forces σ^T itself — the current strategy — to be anywhere near Nash. It is the *averages* that are squeezed. This is the deep reason Deep CFR must train a separate average-policy network, and why judging a mid-training CFR run by its current strategy is a mistake.

Watch out

In an *extensive-form* game (a game tree), “the average strategy” must be defined so that the linearity step in the proof still holds. The correct object weights each round's behavior at an info set by that round's probability of *reaching* the info set under the player's own strategy — averaging the action probabilities naively would break (*). This is exactly why the CFR pseudo-code in the companion volumes accumulates `S[I] += reach[p] * sigma`: that reach factor is not an implementation detail, it is the hypothesis of Theorem B.

5 Ingredient C: from one decision to a whole game tree

Ingredients A and B already prove convergence for games with *one* simultaneous decision (like RPS). Poker, however, is a *tree* of decisions with hidden information. The bridge is the conceptual heart of the CFR paper: define a *local* regret at every info set such that **(i)** each local regret can be driven down by an independent regret matcher (ingredient A applies), and **(ii)** the local regrets jointly control the only thing we actually care about — the global, whole-game regret that ingredient B consumes.

Counterfactual value and counterfactual regret. Fix the current strategy profile σ^t and an infoset I belonging to player i . The *counterfactual value* $v(I, a)$ is the expected payoff to i for playing a at I , where the probability of being at I is weighted only by the *others'* contribution (opponent and chance) — as if player i had steered toward I on purpose. The *immediate counterfactual regret* at I after T rounds is then exactly a Section-2 regret built from these values:

$$R_T^{\text{imm}}(I) = \max_a \sum_{t=1}^T \left(v^t(I, a) - \sum_b \sigma^t(I, b) v^t(I, b) \right).$$

CFR is, by definition, the algorithm that runs one regret matcher per infoset on these quantities.

Because the counterfactual reach weight of the others is at most 1, each $v(I, a)$ has range at most Δ , so Theorem A applies verbatim at each infoset:

$$R_T^{\text{imm}}(I) \leq \Delta \sqrt{AT} \quad \text{for every } I. \quad (\text{A applied locally})$$

What remains is the decomposition: local control implies global control.

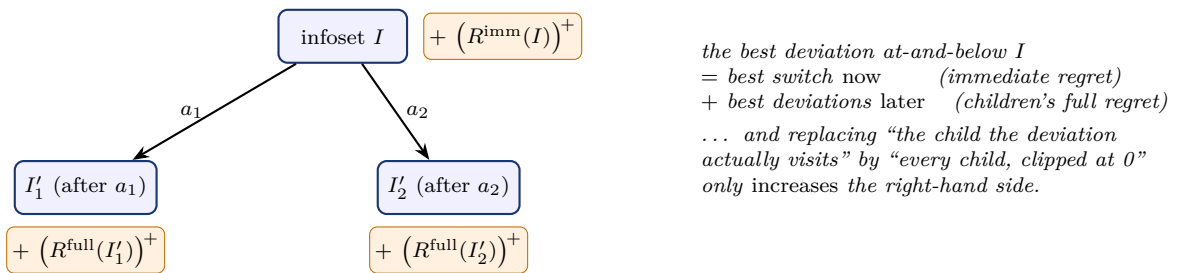
Theorem C (Zinkevich et al. 2007, regret decomposition)

For a player with perfect recall, the whole-game external regret is bounded by the sum of clipped immediate counterfactual regrets:

$$R_T^{\text{game}} \leq \sum_{I \in \mathcal{I}} \left(R_T^{\text{imm}}(I) \right)^+.$$

Proof idea, made precise by induction. The global regret compares your play to the best *deviation strategy* σ' — a full alternative plan that may change the action at *every* infoset at once. The key observation is that any such plan can be charged, infoset by infoset, to local regrets.

Define, for each infoset I of the player, the *at-and-below regret* $R_T^{\text{full}}(I)$: the best gain achievable by deviating at I and at *any of the player's infosets below it*, weighted counterfactually as above. With perfect recall, the player's infosets form a tree (each infoset remembers its own past, so it has a unique parent), and we can induct from the leaves upward:



Formally, one shows for every infoset I :

$$R_T^{\text{full}}(I) \leq R_T^{\text{imm}}(I) + \sum_{I' \in \text{succ}(I)} \left(R_T^{\text{full}}(I') \right)^+,$$

where $\text{succ}(I)$ are the player's next infosets reachable from I . The reasoning: a deviation plan at I first gains whatever its action switch at I alone gains (at most $R_T^{\text{imm}}(I)$, by definition of the immediate regret), and then gains whatever its sub-plan gains below — but each below-gain is counted in that child's own full regret, and clipping at zero lets us include *all* children instead of only the ones the plan steers toward (adding non-negative terms can only help the bound). Taking positive parts of both sides preserves the inequality, and unrolling the recursion from the leaves

(where $R^{\text{full}} = R^{\text{imm}}$) up to the root turns the nested sums into one flat sum over all infosets. At the root, the at-and-below regret *is* the whole-game regret. $\square$

Intuition

Why must the local values be *counterfactual* — weighted by the others’ reach only? Look at where the induction charges a deviation’s gain: to the local regret *at the infoset where the deviation acts*. For that charge to be honest, the local regret must already account for “how often does play arrive here if I try to come here” — i.e. it must not be discounted by the player’s own (possibly deviating) probabilities, which the plan is free to change. Weighting by the others’ reach makes the local ledger deviation-proof, and it is precisely the **reach[1-p]** factor in the pseudo-code of the companion volumes. Perfect recall is what makes the infosets a *tree* so the induction is well-founded — without it, deviations could “merge” paths and gains would be double-counted.

6 The assembly: proving the main theorem

Everything is on the table; the final proof is a four-line chain. Consider player i after T iterations of CFR self-play:

$$\frac{R_T^{\text{game}}}{T} \stackrel{\text{(C)}}{\leq} \frac{1}{T} \sum_{I \in \mathcal{I}} \left(R_T^{\text{imm}}(I) \right)^+ \stackrel{\text{(A)}}{\leq} \frac{|\mathcal{I}| \Delta \sqrt{AT}}{T} = \frac{|\mathcal{I}| \Delta \sqrt{A}}{\sqrt{T}} =: \varepsilon_i.$$

Both players run the same procedure, so both average regrets are at most this ε_i . Feed them to Theorem B:

Main theorem, assembled

After T CFR iterations, $(\bar{\sigma}_1, \bar{\sigma}_2)$ is an ε -Nash equilibrium with

$$\varepsilon \leq \varepsilon_1 + \varepsilon_2 \leq \frac{2 \Delta |\mathcal{I}| \sqrt{A}}{\sqrt{T}} \xrightarrow{T \rightarrow \infty} 0. \quad \blacksquare$$

Each inequality is one ingredient, and each ingredient carries one idea:

step	ingredient	the one idea
game regret $\leq$ sum of local regrets	C (decomposition)	charge deviations infoset-by-infoset
each local regret $\leq \Delta \sqrt{AT}$	A (regret matching)	updates orthogonal to mistakes
small avg. regret $\Rightarrow$ near-Nash	B (folk theorem)	two regret bounds pin the average

7 What the proof promises — and what it does not

Reading the fine print of a theorem is half the value of proving it.

It is the average, never the current strategy. Nothing anywhere constrains σ^T ; the RPS table showed a Paper-spamming current strategy coexisting with a converging average. Practical corollaries: Deep CFR’s deliverable is the policy network trained on the strategy memory, and mid-training sanity checks must target average behavior (or known equilibria, like this project’s stage-1 push/fold oracle).

The bound is worst-case and dimension-aware. ε scales with $|\mathcal{I}|$ (game size), $\sqrt{A}$ (action count), Δ (stake size) and $1/\sqrt{T}$. The $|\mathcal{I}|$ factor is why full-size poker needed sampling and function approximation; a refined analysis replaces $|\mathcal{I}|$ by a smaller game-structure constant, but the $1/\sqrt{T}$ rate is essentially tight for regret-matching-style updates.

Sampling (MCCFR) keeps the guarantee in expectation. External-sampling MCCFR replaces exact counterfactual values with unbiased samples; the same proof skeleton goes through with an extra concentration argument, giving the same $O(1/\sqrt{T})$ rate with high probability (Lanctot et al. 2009). Unbiasedness of the sampled regrets is the load-bearing property — this is why implementation bugs that bias the sampling distribution (e.g. the strategy-memory and action-indexing bugs found in this project’s trainer) genuinely break convergence rather than merely slowing it.

Function approximation adds an error floor. Deep CFR replaces the per-inforeset regret tables with a regression network. Brown et al. (2019) show the same convergence statement with an additional term proportional to the regression error: the average strategy converges to an $O(\varepsilon_{\text{approx}})$ -Nash rather than an exact one. The $1/\sqrt{T}$ part still requires the networks to be *trained well each iteration* — a vivid lesson from this repository, where a mild `weight_decay` quietly inflated $\varepsilon_{\text{approx}}$ to the point of freezing the strategy entirely.

Two players, zero-sum, perfect recall — all three matter. Theorem B’s proof used zero-sum cancellation; with three or more players low regret no longer implies an equilibrium (this is why Pluribus, in volume two, abandons the theoretical target). Theorem C’s induction used perfect recall to make the infosets a tree.

Better constants exist, same skeleton. CFR^+ (clip cumulative regrets, alternate updates) and Linear CFR (weight round t by t) both admit the same style of proof with strictly better constants — the potential argument survives clipping (Lemma 2 is exactly what is needed) and survives weighting (sum tr_t instead of r_t ; the orthogonality is unchanged). Empirically they converge several times faster, which is why this project’s trainer uses Linear CFR.

8 Further reading

- Hart, Mas-Colell (2000). *A simple adaptive procedure leading to correlated equilibrium* — regret matching and (a generalization of) Theorem A.
- Zinkevich, Johanson, Bowling, Piccione (2007). *Regret Minimization in Games with Incomplete Information* — counterfactual values, Theorems C and the assembled bound.
- Lanctot, Waugh, Zinkevich, Bowling (2009). *Monte Carlo Sampling for Regret Minimization in Extensive Games* — the sampled variants and their probabilistic guarantees.
- Tammelin (2014); Bowling et al. (2015). CFR^+ and the essential solving of limit hold'em.
- Brown, Sandholm (2019). *Solving Imperfect-Information Games via Discounted Regret Minimization* — Linear/discounted CFR and their proofs.
- Brown, Lerer, Gross, Sandholm (2019). *Deep Counterfactual Regret Minimization* — the approximation-error version of the theorem.
- Cesa-Bianchi, Lugosi (2006). *Prediction, Learning, and Games* — the textbook home of regret minimization and the folk theorem.
- Volume one: *Deep CFR, explained*; volume two: *After Deep CFR* (`docs/deep_cfr_tutorial/`, `docs/modern_poker_ai/`).